From Anonymous Shapes to Named Places

A Tool for Braille and Place-Semantic Annotation of Tactile Maps

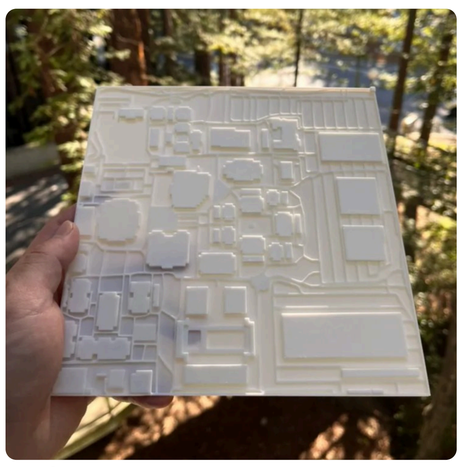
Li Liu[1], Ashmita Dua[1], Jiaming Qu[2,*], David T. Lee[1], Leilani H. Gilpin[1]
[1]University of California, Santa Cruz [2]Amazon ASSETS '26 · Demo 1266
Contact: lliu112@ucsc.edu

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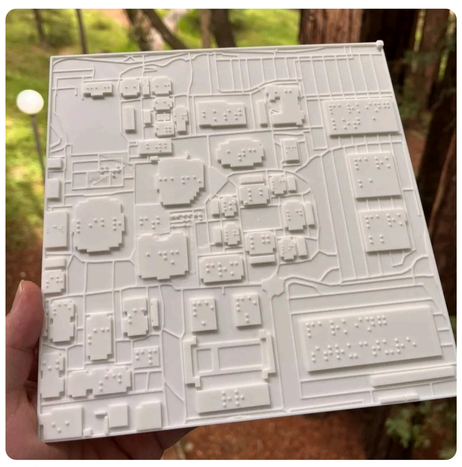
Try the tool

On a tactile map, a building under the finger is an anonymous shape.

Our tool adds Braille labels to tactile maps that already exist, so each shape on the print carries its own name.



Before · no labels



After · with Braille



Why we need it · A tactile map gives the *where*. The *what* is hard to get.

Hard to read · for the reader

Hard to make · for the helper

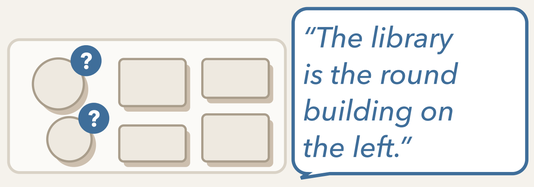
1



Felt, but not named

On a Touch Mapper† print, every building is a shape with no name.

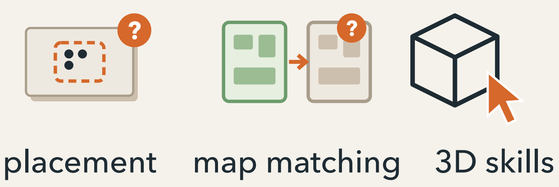
2



Described, but hard to match

A spoken description is hard to tie to the right shape.

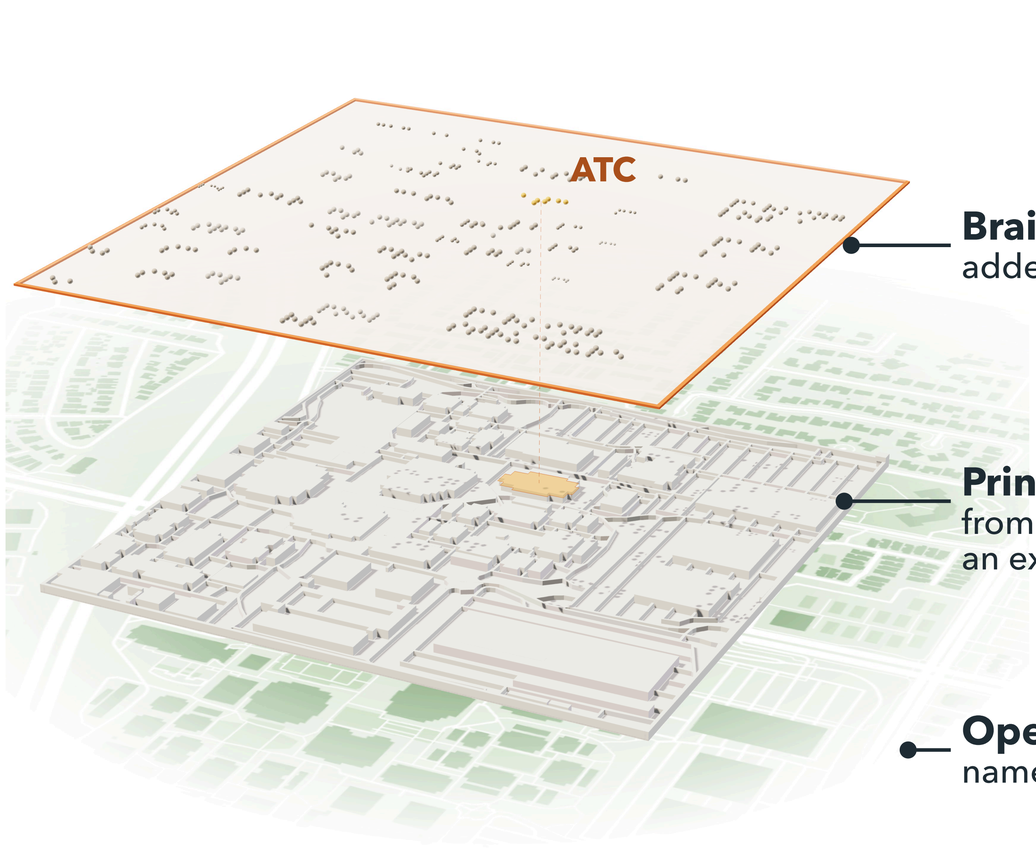
3



Hard to label by hand

Each label means matching the print to a map, then editing it in 3D software.

What our tool does · The 3D model and its map, opened side by side and linked

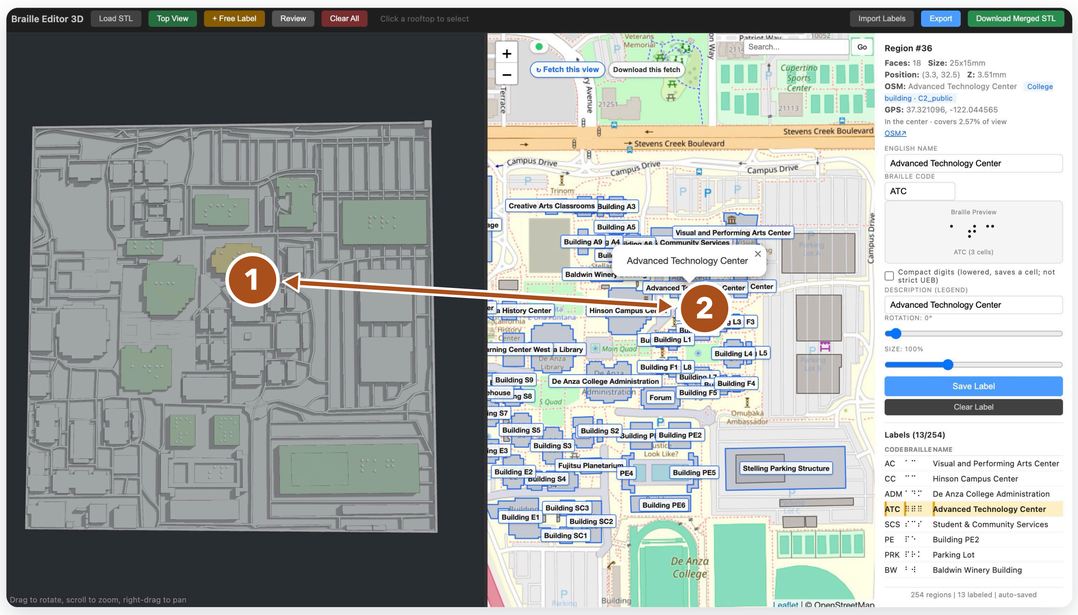


Braille layer
added with our tool

Printed base
from Touch Mapper, an existing tool

OpenStreetMap
names and places

3D model from Touch Mapper · **OpenStreetMap** · **Label**



1 Load the model. The same area opens on OpenStreetMap.

2 Click it on both sides. Name and Braille fill in, no typing.

3 Review, then export. The Braille attaches to the model base.

What comes next · From one labeled print to a shared Braille map of the world

Today: one print

- Braille on the print
- Braille legend
- Print-ready 3D model

Each label is saved

a shape on the print ↓ a place on OpenStreetMap

Next: a shared, reusable map

- Reused for the next print
- Audio, symbols, AI agents
- Built together by everyone
- Shared layer over OpenStreetMap

Project page leolee7.github.io/#project-tactile-annotation · **Paper** doi.org/10.1145/3797867.3841235

†Touch Mapper, an existing tool that makes tactile maps from OpenStreetMap: touch-mapper.org
*Work unrelated to position at Amazon.

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